

Acceleration of One-Head Transformer Attention on a Xilinx Alveo U55C FPGA

Samarth Bangalore Sudharshan, Rijul Wankhade, Om, and Satyarth
CEN571: Reconfigurable Computing
Arizona State University

Abstract—Transformer inference is increasingly important for edge and local deployment, but attention layers remain expensive because their score matrix grows quadratically with sequence length. This project investigates whether a specialized FPGA data path can accelerate the core attention computation used by a TinyLlama-derived model. The implemented system targets a Xilinx Alveo U55C and offloads one attention head, beginning from RoPE-applied integer-quantized query and key tensors and a floating-point value tensor. The verified hardware path computes INT8 query-key scores, applies causal masking and dequantization scaling, performs full-row softmax, and computes $\text{softmax}(QK^T)V$ to produce a one-head floating-point attention output. The design is intentionally narrower than full TinyLlama inference: it does not implement Q/K/V projection, RoPE generation, decoder integration, or model-level token generation. The final fused design passes local C++ checks, Vitis HLS C simulation and synthesis, XRT hardware emulation, and real U55C execution for synthetic sequence lengths $S = 8, 64, 128, 256, 512$ and TinyLlama-derived vector directories at $S = 16$ and $S = 64$. On real hardware, the fused FPGA path runs the synthetic $S = 512$ one-head attention case in 60.328 ms, compared with a 12.329 ms CPU NumPy baseline and a 0.246 ms RTX 3050 Laptop GPU PyTorch baseline. These results show a correct and presentable FPGA attention block with clean post-route timing closure, but also show that the current staged architecture is dominated by repeated kernel launches and host-visible HBM staging. The work establishes a validated baseline and identifies the next required optimizations: device-resident data flow, reduced launch count, online softmax fused with value accumulation, and parallel execution across query blocks or heads.

Keywords—FPGA, transformer attention, TinyLlama, Xilinx Alveo U55C, HLS, XRT, HBM, softmax, quantization.

I. INTRODUCTION

Large language models are moving from cloud-only services toward local and edge deployments, where privacy, availability, and latency constraints make remote execution less attractive. Edge LLM systems must process user prompts locally, respond within a tight latency budget, and operate under more limited power and thermal envelopes than data-center GPUs. These requirements motivate hardware acceleration techniques that can exploit the structure of transformer computation rather than relying only on general-purpose processors.

The attention mechanism is one of the defining operations in transformer models. For a sequence of S tokens, each attention head computes a score matrix QK^T , applies a causal mask and scaling, converts each score row to probabilities with softmax, and uses those probabilities to form a weighted sum of values. Even for one head, the score matrix is $S \times S$, so storage and compute grow quickly as the sequence length increases. In

a full model such as TinyLlama, this operation is repeated across multiple layers and multiple attention heads. This regular structure makes attention a natural candidate for FPGA acceleration, especially when combined with low-precision arithmetic, tiling, on-chip buffering, and high-bandwidth memory.

This project studies the FPGA offload of a focused attention block rather than a full language model. The goal is to understand what is required to map TinyLlama-style attention onto the Xilinx Alveo U55C, verify the result on real hardware, and measure where the first implementation succeeds or falls short. The selected hardware boundary begins after the model has already generated RoPE-applied query and key tensors and a value tensor. Query and key are quantized to INT8 with per-tensor scale factors, while value and final attention output remain FP32. The FPGA computes the one-head attention result:

$$\text{attn.out} = \text{softmax}\left(\frac{QK^T}{\sqrt{d_h}} + M\right)V,$$

where $d_h = 64$ is the head dimension and M is the causal mask.

The project evolved through four required tracks. Track A made the attention score pipeline work across real sequence lengths using tiling, full-row softmax, double buffering for the pre-softmax pass, and finally a fused score-mask-scale kernel. Track B added the value weighted-sum stage so the FPGA produces final one-head attention output rather than only attention probabilities. Track C replaced purely synthetic inputs with TinyLlama-derived Q/K/V vectors. Track D collected CPU, GPU, FPGA, timing, utilization, and verification results. The resulting design is a correct one-head attention accelerator, not a complete TinyLlama runtime. This distinction matters because the reported times are attention-block latencies, not generated-token throughput.

The main contribution of the project is therefore twofold. First, it provides a complete verified U55C implementation of a tiled one-head attention path, including Python reference generation, HLS kernels, XRT host orchestration, HBM bank placement, hardware emulation, real-card validation, and post-route reports. Second, it exposes the architectural bottleneck in a straightforward staged implementation: although the FPGA kernels are correct and timing closes, the system remains slower than CPU and GPU baselines because it launches many small kernels and materializes intermediates in host-visible HBM buffers.

II. BACKGROUND AND PRIOR WORK

A. Edge LLM Motivation

Running language models near the user has different requirements from running them in a centralized GPU cluster. A local assistant, industrial controller, or embedded analysis system may need to keep prompts private, continue operating when a network connection is unavailable, and stay within a fixed power budget. At the same time, the system must still perform the transformer operations that dominate modern language models. The result is a hardware problem: the accelerator must provide enough throughput and low enough latency without the thermal and cost assumptions of a large server GPU.

FPGAs are attractive in this setting because they can be specialized after deployment. If a model uses fixed head dimensions, fixed quantization formats, or a known maximum sequence length, an FPGA design can harden those choices into the data path. Unlike a CPU, it can expose many operations in parallel. Unlike a GPU, it can avoid some general-purpose scheduling overhead and can use custom memory layouts. However, these benefits only appear when the design is architecturally matched to the workload. A naive staged FPGA implementation can easily lose to software if the host must launch hundreds or thousands of small kernels and move intermediate matrices through external memory after every stage. This project demonstrates both sides of that tradeoff: a correct FPGA attention block was built and verified, but the first staged implementation is not yet the performance point needed for an edge LLM accelerator.

B. Transformer Attention

The transformer architecture introduced self-attention as a replacement for recurrent sequence processing [1]. In a causal decoder model, each token is allowed to attend only to itself and previous tokens. For one attention head, the input representation is projected into query, key, and value matrices:

$$Q, K, V \in \mathbb{R}^{S \times d_h}.$$

The unnormalized attention scores are:

$$L = QK^T,$$

where $L \in \mathbb{R}^{S \times S}$. The scaled and masked logits are:

$$\hat{L}_{ij} = \begin{cases} L_{ij}/\sqrt{d_h}, & j \leq i, \\ -\infty, & j > i. \end{cases}$$

Softmax is applied independently to each row:

$$P_{ij} = \frac{e^{\hat{L}_{ij} - m_i}}{\sum_{k=0}^{S-1} e^{\hat{L}_{ik} - m_i}}, \quad m_i = \max_j \hat{L}_{ij}.$$

Finally, attention output is:

$$O = PV.$$

The subtraction of m_i is the standard numerically stable softmax form.

TinyLlama uses hidden dimension 2048 and 32 query heads, which gives a head dimension of 64 [2]. The project focuses on one selected head. TinyLlama also uses grouped-query attention with fewer key/value heads than query heads; the export scripts map the selected query head to the corresponding key/value head before producing vectors for FPGA verification.

C. FPGA Suitability

FPGAs are well suited to regular, streaming, and tiled data paths. They allow custom arithmetic width, explicit memory banking, loop pipelining, and fine-grained parallelism. For attention, the QK^T stage is a dense dot product over a fixed head dimension, while the PV stage is another matrix product over the sequence axis. Both operations can be tiled to fit local buffers and mapped to DSP-heavy datapaths.

The Alveo U55C provides HBM, multiple SLRs, and a Vitis/XRT software stack for kernel compilation and host control [3]–[6]. In this project, XRT is responsible for loading the xclbin, allocating buffer objects, syncing data between host and device, launching kernels, and reading back results. Vitis HLS is used to compile C++ HLS top functions into FPGA kernels.

D. Prior Acceleration Approaches

Prior work on neural network acceleration often focuses on reducing precision, reusing on-chip data, and fusing operations to avoid external memory traffic. Transformer accelerators additionally try to avoid materializing the full attention probability matrix, because it requires $O(S^2)$ storage. Online softmax algorithms address this by streaming over key/value blocks while maintaining a row maximum, normalization sum, and value accumulator. This project's verified baseline does not yet implement online softmax, but the optimization checklist identifies it as the most important future step after device-resident buffers and launch-count reduction.

The project therefore follows a staged design methodology rather than jumping directly to a monolithic online-attention kernel. Each stage is verified first: score generation, mask and scale, full-row softmax, and value accumulation. The benefit of this approach is observability. Intermediate tensors can be compared against Python references, which makes debugging much easier. The drawback is that a staged design materializes intermediate data and exposes more control work to the host. The final report results should be interpreted with that tradeoff in mind.

III. PROJECT DESCRIPTION

A. Scope

The implemented system accelerates one attention head from a TinyLlama-derived path. The input to the FPGA-side computation is:

$$Q_{\text{int8}}, K_{\text{int8}}, V_{\text{fp32}},$$

where Q and K already include rotary positional embedding in the exported software flow. The output is:

$$O_{\text{fp32}} \in \mathbb{R}^{S \times 64},$$

the attention output for one head. The design does not produce text, logits over the vocabulary, or a complete transformer-layer output.

The out-of-scope components are Q/K/V projection GEMMs, RoPE generation, layer normalization, feed-forward layers, key/value cache management, decoder-layer integration, and full TinyLlama token generation. Keeping these components out of scope made the project measurable within a semester-scale hardware design effort while still preserving the mathematically important part of attention.

The design also focuses on prefill-style full-sequence attention rather than incremental decode with a key/value cache. In a decode workload, each new token uses one query row and attends over cached keys and values. In the implemented test cases, the full $S \times S$ attention pattern is evaluated for a saved sequence. This is appropriate for validating the core matrix operations and for comparing synthetic sequence lengths, but a deployed LLM runtime would need additional logic for cache layout, streaming token inputs, and interaction with the surrounding decoder layer.

B. Data Types and Quantization

The query and key tensors are quantized using per-tensor symmetric INT8 scales. If q_s and k_s are the scale factors, the approximate dequantized values are:

$$Q_{fp32} \approx q_s Q_{int8}, \quad K_{fp32} \approx k_s K_{int8}.$$

The raw dot product is accumulated in INT32:

$$R_{ij} = \sum_{d=0}^{63} Q_{int8}[i, d] K_{int8}[j, d].$$

The fused score-mask-scale kernel then computes:

$$\hat{L}_{ij} = \begin{cases} R_{ij} q_s k_s / 8, & j \leq i, \\ -10^9, & j > i, \end{cases}$$

where $1/\sqrt{64} = 1/8$. The implementation uses a large negative finite mask value rather than literal negative infinity to keep the kernel simple and deterministic. Softmax, value, and attention output use FP32 in the verified design.

C. Tiling

The hardware tile shape is fixed around the head dimension:

$$\begin{aligned} Q_{\text{tile}} &: 8 \times 64, \\ K_{\text{tile}} &: 64 \times 64, \\ \text{score}_{\text{tile}} &: 8 \times 64. \end{aligned}$$

For sequence length S , the number of score tiles per head is:

$$N_{\text{tiles}} = \left\lceil \frac{S}{8} \right\rceil \left\lceil \frac{S}{64} \right\rceil.$$

The implementation handles non-full tiles through active row and column counts. Inactive rows or columns are padded and ignored rather than discarded.

For $S > 64$, softmax cannot be applied independently to each 8×64 score tile. Each query row must be normalized

TABLE I
TILE COUNTS FOR THE TESTED SYNTHETIC SEQUENCE LENGTHS.

S	Q chunks	K chunks	Score tiles
8	1	1	1
64	8	1	8
128	16	2	32
256	32	4	128
512	64	8	512

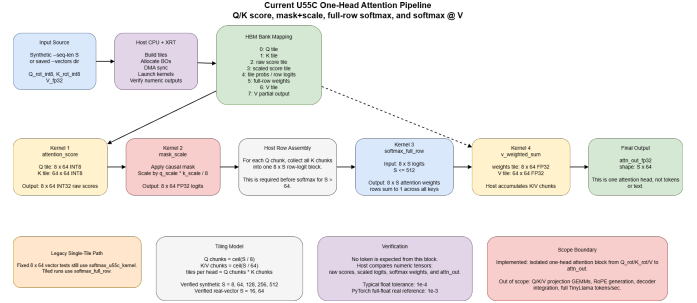


Fig. 1. Frozen one-head attention architecture. The verified master design uses host/XRT orchestration, HBM banked buffers, fused score-mask-scale, full-row softmax, and a value weighted-sum stage.

across all S keys. Therefore, the host assembles full $8 \times S$ logit rows from all key chunks, launches a full-row softmax kernel, slices the resulting probability rows into 8×64 tiles, and passes those tiles to the value weighted-sum kernel.

For example, at $S = 512$, the FPGA processes 64 Q chunks and 8 K chunks per head. The score stage therefore launches 512 tile computations. Full-row softmax launches once per Q chunk, for 64 more launches. The V weighted-sum stage launches once per Q/K chunk, adding another 512 launches. This yields 1088 kernel launches for one head. That launch count is the central performance problem observed in the final results.

D. Design Evolution Across Project Tracks

The project was organized around four required tracks. This structure helped separate correctness, data sourcing, and measurement work.

Track A focused on making the score path usable at realistic sequence lengths. The initial implementation demonstrated a single attention-score tile. That was useful for validating the arithmetic, but it did not handle full $S \times S$ attention. Track A added full-sequence tiling in the Python reference and in the host application. It also exposed the important softmax issue: for multiple K chunks, softmax must normalize across the full row, not within each tile. The full-row softmax kernel was therefore added before long-sequence correctness could be claimed. Track A also introduced host-side double buffering in the first pass and eventually fused score, causal mask, and scale into one kernel.

Track B extended the output contract. Before Track B, the FPGA could verify attention weights, but the attention block in a transformer does not stop at weights. It multiplies

those weights by V to produce the attended output. Track B added the Python reference for `softmax@V`, implemented the HLS value weighted-sum kernel, updated host integration, and changed verification so that final $S \times 64$ attention output is checked. This made the FPGA demo much closer to a real transformer attention sub-block.

Track C replaced purely synthetic data with TinyLlama-derived vectors. This track first verified that the local Python environment could load TinyLlama, then extracted Q , K , and V tensors from one attention layer, applied the head-mapping needed for grouped-query attention, quantized Q and K , and exported saved-vector directories. Track C is the reason the project can claim TinyLlama-derived input validation rather than only deterministic toy data.

Track D created the measurement story. It added CPU and GPU baselines, real U55C timing sweeps, utilization reports, and a discussion of what the results do and do not mean. Track D is especially important because it prevents an overly optimistic conclusion. The FPGA is functionally correct, but the first staged architecture is not yet faster than the one-head CPU baseline. That result drives the optimization checklist and the future-work direction.

IV. METHODOLOGY

A. Reference Model and Vector Export

The software reference is written in Python and serves three purposes. First, it defines the expected numerical result for each stage of the attention pipeline. Second, it exports deterministic synthetic vectors for repeatable unit tests and hardware tests. Third, it extracts TinyLlama-derived $Q/K/V$ vectors from a real model path for saved-vector verification.

The synthetic path generates deterministic Q , K , and V tensors for sequence lengths $S = 8, 64, 128, 256, 512$. The Python reference computes raw INT32 scores, masked and scaled logits, full-row softmax probabilities, and final attention output. The TinyLlama path loads TinyLlama/TinyLlama-1.1B-Chat-v1.0, captures $Q/K/V$ at a selected attention layer, applies the same grouped-query head mapping used by the model, quantizes Q and K , and writes vector directories under `sim/`. The verified real-vector directories include a legacy single-tile case and full-sequence $S = 16$ and $S = 64$ cases.

The reference scripts intentionally produce simple text files rather than a binary-only format. This made the files usable by Python, local C++ testbenches, HLS C simulation, and the XRT host application. The main exported tensors are Q , K , scaled logits, softmax probabilities, V , and final attention output. The single-tile directories also preserve raw and masked score references for debugging. In the fused hardware path, raw scores are internal to the kernel and are no longer written back to HBM, but keeping the raw-score reference helped validate that fusion did not change the mathematical contract.

The TinyLlama extraction flow had two checks. First, the setup script loaded the tokenizer and model and ran a short forward pass to confirm that the software environment

could execute TinyLlama. Second, the $Q/K/V$ extraction script registered a pre-hook on a selected attention layer, mirrored the model's projection and RoPE behavior, and compared the selected-head attention result against PyTorch scaled-dot-product attention. The default extraction run reported an SDPA maximum difference of approximately 3.7×10^{-9} , which gave confidence that the saved vectors represented the intended model subgraph before quantization.

After extraction, Q and K were quantized per tensor. A representative run reported Q scale 4.8984×10^{-2} , K scale 1.7432×10^{-2} , Q reconstruction maximum error 2.45×10^{-2} , K reconstruction maximum error 8.71×10^{-3} , and dequantized score maximum error 2.66×10^{-2} . These numbers are not model accuracy measurements; they only quantify the local numerical error introduced by the INT8 Q/K representation used for this one-head accelerator.

B. HLS Kernel Design

The verified fused design contains four main kernels in the real-card attention path:

- 1) `score_mask_scale_u55c_kernel`: reads packed INT8 Q and K tiles, computes QK^T , applies causal mask and scale, and writes FP32 logits directly.
- 2) `softmax_full_row_u55c_kernel`: normalizes up to eight full query rows over $S \leq 512$ keys.
- 3) `softmax_u55c_kernel`: preserved for the legacy single-tile softmax path.
- 4) `v_weighted_sum_u55c_kernel`: computes one partial 8×64 contribution of PV from an 8×64 probability tile and a 64×64 V tile.

The fused score-mask-scale kernel replaces the earlier staged score GEMM plus mask/scale kernels. This removes a raw-score HBM round trip and one kernel launch in the pre-softmax path. The V weighted-sum kernel remains staged: for each Q chunk, the host launches it once per K/V chunk and accumulates partial outputs into final $S \times 64$ attention output.

The design history is useful for understanding the final architecture. The earliest score kernel used a narrow external memory interface and spent most of its estimated latency loading K . Widening Q , K , and score traffic to packed 64-bit words reduced the score-kernel latency estimate from roughly 5153 cycles to roughly 1312 cycles. This showed that the first bottleneck was not only the number of multiply-accumulate operations, but also how data was presented to the kernel. Later, causal mask and score scale were merged so the design no longer needed two separate post-score kernels. Finally, score generation, masking, and scaling were fused into `score_mask_scale_u55c_kernel`, which writes FP32 logits directly.

The full-row softmax kernel was added after the project moved beyond single-tile sequence lengths. A tile-local softmax is correct only when all valid keys fit inside one 8×64 tile. For $S > 64$, normalizing each tile independently would make each tile row sum to one and would over-count probability mass across K chunks. The full-row kernel therefore accepts

TABLE II
HBM BANK MAPPING IN THE FUSED VERIFIED PATH.

Data buffer	HBM bank
Q tile	HBM[0]
K tile	HBM[1]
Fused scaled logits	HBM[3]
Tile probability / full-row logits	HBM[4]
Full-row probability / V weights	HBM[5]
V chunk	HBM[6]
V partial output	HBM[7]

an $8 \times S$ logit block and performs a stable row-wise softmax across all keys. The current maximum supported full-row length is $S = 512$, matching the synthetic sweep used for evaluation.

The HLS design uses loop pipelining, unrolling, and array partitioning rather than a hand-written systolic array. The dot-product loops are unrolled to expose parallel multipliers, local arrays are partitioned so the pipeline has enough read ports, and memory interfaces are mapped to HBM banks through the Vitis link configuration.

This means the design should not be described as a systolic array. A systolic array would pass data rhythmically through a grid of processing elements. The implemented kernels instead rely on HLS scheduling: data is loaded into local arrays, loops are pipelined, and selected inner loops are unrolled so multiple multiply-accumulate operations occur in parallel. This approach is easier to iterate in C++ HLS and sufficient for the first verified design, but it leaves room for future hand-optimized or more deeply streaming architectures.

C. XRT Host Flow and HBM Mapping

The host application is written with the XRT C++ API. It loads the xclbin, creates kernels, allocates buffer objects, writes vectors, synchronizes buffers to the device, launches kernels, waits for completion, synchronizes results back, and compares outputs against reference tensors. The main XRT calls are `xrt::device`, `device.load_xclbin`, `xrt::kernel`, `xrt::bo`, `bo.write`, `bo.sync`, `kernel invocation`, `run.wait`, and `bo.read`.

Table II lists the fused design’s HBM bank mapping. The earlier raw score bank is no longer used by the fused pre-softmax path because raw scores are immediately masked and scaled inside the fused kernel.

The host is also responsible for sequencing dependencies between kernels. For each Q chunk, it launches the fused pre-softmax kernel across K chunks, gathers the resulting 8×64 logit tiles into an $8 \times S$ row block, launches full-row softmax, slices the probabilities back into K-sized blocks, launches V weighted-sum kernels, and accumulates the partial 8×64 outputs. This makes the data flow easy to verify because every major intermediate is visible to the host. It also explains why the final design has substantial overhead: the host remains in the middle of the attention data path.

TABLE III
FPGA PLATFORM AND SOFTWARE ENVIRONMENT.

Component	Details
FPGA card	AMD/Xilinx Alveo U55C
FPGA device	xcu55c-fsvh2892-2L-e
Platform	xilinx_u55c_gen3x16_xdma_3.202210.1
Tools	Vitis/Vivado/Vitis HLS 2022.2
Runtime	XRT 2022.2
Host compiler	g++ with C++17
Kernel clock	300 MHz target
HBM clock	450 MHz

D. Verification Flow

Verification was performed in layers. Python reference checks validated the math for full-sequence tiling at $S = 8, 64, 128, 256, 512$. Local C++ testbenches validated individual HLS kernels against exported files. Vitis HLS C simulation and synthesis were run for the kernel implementations. XRT hardware emulation validated the compiled xclbin and host application before running on the real card. Finally, the hardware bitstream was executed on a U55C and compared against software references.

The project did not use Vitis HLS C/RTL co-simulation. Instead, the verification chain combined Python references, local C++ tests, HLS `csim`, `csynth`, XRT `hw_emu`, and real-card XRT execution. The expected pass signal for full attention runs is:

```
Attention output verification PASSED
```

followed by:

```
XRT chain verification PASSED.
```

Floating-point comparisons use a tolerance of 10^{-4} for most synthetic and intermediate comparisons, with a larger 10^{-3} tolerance for some TinyLlama-derived final attention outputs where quantization and PyTorch reference differences are larger.

The verification flow deliberately separates correctness from performance. A kernel can pass HLS simulation and still fail to place, route, or perform well after integration. Conversely, hardware emulation can validate API-level behavior and final output, but its runtime is simulator dominated and should not be used as a performance estimate. Real-card timing is therefore reported only from U55C hardware runs with `XCL_EMULATION_MODE` unset.

E. Experimental Setup

The FPGA experiments used Vitis/Vivado 2022.2, XRT 2022.2, and the U55C platform `xilinx_u55c_gen3x16_xdma_3.202210.1`. The card shell reported `xilinx_u55c_gen3x16_xdma_base_3`. The real-card runs used device index 0. Hardware builds produced an xclbin bitstream, which the host loaded through XRT.

CPU baselines were measured with the repository’s NumPy/Python benchmark scripts. GPU baselines were

TABLE IV
VERIFICATION SUMMARY FOR THE FUSED ONE-HEAD ATTENTION
PATH.

Stage	Result
Python reference	Full tiling check passed for $S = 8, 64, 128, 256, 512$.
Local C++ benches	Score, fused score-mask-scale, softmax, full-row softmax, and V weighted sum passed.
Vitis HLS	Relevant kernels passed <code>csim</code> and <code>csynth</code> .
XRT <code>hw_emu</code>	Fused vector and sequence tests passed.
Real U55C	Synthetic $S = 8, 64, 128, 256, 512$ and real-vector $S = 16, 64$ passed through final attention output.

measured separately on a Windows laptop with PyTorch 2.11.0+cu128, CUDA 12.8, and an NVIDIA GeForce RTX 3050 Laptop GPU. The GPU baseline is useful as a PyTorch comparison point, but it is not a same-host measurement with the Linux/U55C system. This distinction is important when interpreting speedup tables.

The FPGA hardware build produces more than a single executable artifact. The checked-in report set includes xclbin information, link summaries, system diagrams, routed timing summaries, full-design utilization, per-kernel utilization, and SLR utilization. These artifacts are used in the Results section to support timing-closure and resource claims. The paper intentionally uses summarized tables and diagrams rather than large tool screenshots, because the assignment asks for an IEEE-style report rather than a collection of build logs.

V. RESULTS AND DISCUSSION

A. Functional Verification

Table IV summarizes the main verification outcomes. The final fused design was validated with both synthetic inputs and saved TinyLlama-derived vectors. Synthetic sequence lengths cover multiple tile shapes, including multi-K-chunk cases. Real TinyLlama saved-vector directories currently cover $S = 16$ and $S = 64$, plus the legacy single-tile case.

The most important functional result is that the FPGA output is verified at the final attention-output tensor, not only at intermediate scores. Earlier stages of the project verified attention weights, but Track B extended the design so that `softmax@V` is also computed and checked. This makes the result more representative of the actual attention block used by a transformer layer. At the same time, the output remains one-head $S \times 64$ data, not generated text or model logits over the vocabulary.

B. Latency Results

Table V shows synthetic timing results. The staged FPGA baseline uses separate pre-softmax stages, while the fused FPGA design combines score computation, masking, and scaling. The fused path improves all synthetic sequence lengths

relative to the staged FPGA baseline, with a 1.21x improvement at $S = 512$. However, the FPGA remains slower than the CPU and GPU baselines for this isolated one-head workload.

Table VI shows real TinyLlama-derived saved-vector results. The fused design improves the full-sequence $S = 16$ and $S = 64$ real-vector runs compared with the staged FPGA baseline. The legacy single-tile `real_tinyllama_tile` case is slower after fusion, which is treated as a small-run overhead/noise case rather than the representative regime. The larger real-vector $S = 16$ and $S = 64$ cases are more meaningful because they exercise the tiled full-sequence path.

The synthetic sweep shows that total FPGA latency grows with sequence length, as expected from the S^2 score matrix and the increasing number of tile launches. The fused path is consistently better than the staged FPGA baseline for synthetic tests because the pre-softmax chain is shorter. However, the CPU/fused-FPGA ratio remains below 1.0 for every tested sequence length, meaning the CPU baseline is faster for this isolated workload. At $S = 512$, the FPGA would need roughly a 5x end-to-end improvement to beat the measured CPU baseline. That level of improvement is unlikely to come from small HLS pragma tuning alone.

The measured GPU times are nearly flat across the synthetic sequence lengths in the table. This is partly because the benchmarked one-head problem is small relative to the GPU’s available compute and library overheads. The GPU is not being saturated in the same way it would be for a full model or batched workload. Nevertheless, it provides a useful reference: for small dense tensor operations, optimized GPU software is a very strong baseline.

The comparison also highlights why the evaluation scope must be stated carefully. A full TinyLlama layer would compute all heads, apply output projection, and continue through the rest of the decoder block. The reported tables isolate only one attention head after Q/K/V are already prepared. This is the right scope for testing the FPGA attention block, but it is not a claim about complete model latency. In a complete accelerator, some fixed host overheads could be amortized across more heads or fused with neighboring model operations; the present staged design does not yet do that.

C. Utilization and Timing Closure

The final routed fused xclbin closes timing on the U55C. The post-route timing summary reports $WNS = 0.003$ ns, $TNS = 0$, $WHS = 0.009$ ns, and no failing setup or hold endpoints. The HLS kernels target approximately 300 MHz, and the platform reports HBM at 450 MHz.

Table VII summarizes post-route utilization. The full routed design, including the platform, uses 14.70% CLB LUTs, 9.72% CLB registers, 15.35% Block RAM tiles, 0.21% URAM, and 4.53% DSP. The user kernels are therefore not capacity-limited in the fused baseline. The largest kernel is `v_weighted_sum_u55c_kernel`, which uses 320 DSP and dominates the user-kernel DSP count.

The utilization numbers show that the fused baseline is not close to filling the FPGA. This is encouraging for future

TABLE V
SYNTHETIC ONE-HEAD ATTENTION LATENCY. LOWER IS BETTER.

S	Tiles	CPU ms	GPU ms	Staged FPGA ms	Fused FPGA ms	CPU/Fused FPGA
8	1	0.0322	0.3349	0.803	0.296	0.11x
64	8	0.2920	0.2507	2.864	2.364	0.12x
128	32	2.5009	0.2998	6.992	5.484	0.46x
256	128	3.9523	0.2726	20.570	17.307	0.23x
512	512	12.3290	0.2461	72.941	60.328	0.20x

TABLE VI
REAL TINYLLAMA SAVED-VECTOR LATENCY.

Case	S	CPU ms	Staged ms	Fused ms
real.tinyllama.tile	tile	n/a	0.258	0.635
real.tinyllama.s16	16	0.0426	1.488	0.569
real.tinyllama.s64	64	0.1523	3.094	2.004

TABLE VII
ROUTED UTILIZATION FOR THE FUSED STEP 5 XCLBIN.

Resource	Used	Available	Util.
CLB LUTs	191,648	1,303,680	14.70%
CLB registers	253,388	2,607,360	9.72%
BRAM tiles	309.5	2,016	15.35%
URAM	2	960	0.21%
DSP	409	9,024	4.53%

parallelization, because additional compute units or a more integrated online-attention kernel may fit. It also means that the current poor speedup is not due to lack of raw FPGA resources. Instead, the problem is that the resources are not kept busy enough. The host must repeatedly launch small kernels and synchronize buffers, so the data path has idle time between useful compute regions.

The V weighted-sum kernel is the largest individual kernel in the verified baseline. It consumes 320 DSP, while the fused score-mask-scale kernel consumes 67 DSP and the softmax kernels are much smaller. This aligns with the mathematics: after softmax, each 8×64 probability tile is multiplied by a 64×64 V tile, producing 512 output elements through many floating-point multiply-accumulate operations. Future designs must either make this stage more resident and less host-driven or fuse it with online softmax so the probability matrix is never staged as a separate full tensor.

SLR utilization is also useful for interpreting future designs. The verified fused baseline closes timing and uses a modest fraction of the full device, but larger experimental kernels can still fail if their placement is concentrated in one region or creates too many routing/control-set constraints. Therefore, future optimization work should not look only at total DSP percentage. It must also consider pblock capacity, SLR placement, routing pressure, and whether the datapath is too wide to route cleanly at the target clock.

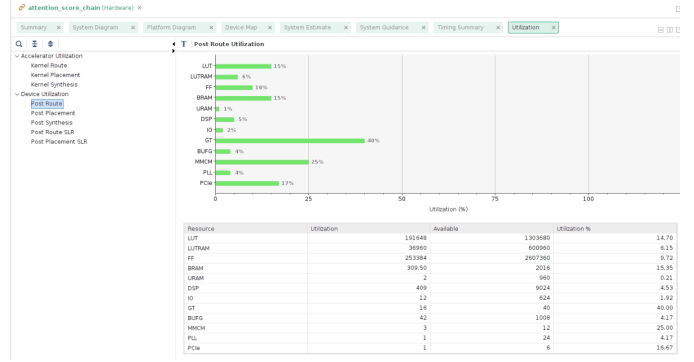


Fig. 2. Post-route utilization summary captured from the generated reports. The fused baseline has significant remaining resources, but the staged architecture still suffers from launch and memory-staging overhead.

D. Discussion of Bottlenecks

The primary lesson from the results is that kernel correctness and timing closure are not sufficient for end-to-end acceleration. At $S = 512$, the fused design still performs: 512 score-mask-scale launches, 64 softmax launches, and 512 V launches, or 1088 total launches. The host also assembles full rows, slices probability tiles, moves V chunks, and accumulates partial outputs. The U55C has substantial remaining resources, but the current orchestration exposes too much intermediate state to host-visible buffers and pays many launch/synchronization costs.

Step 5 fusion helps because it removes one kernel launch and one raw-score HBM round trip from the pre-softmax path. It does not address the deeper issue that softmax and value accumulation remain separate staged passes. This explains why the fused design improves FPGA-to-FPGA comparisons but still does not beat the CPU baseline for the isolated one-head workload.

The GPU baseline is especially fast for these small cases because PyTorch and CUDA execute the matrix operations with highly optimized libraries and low effective launch overhead relative to the work. The CPU baseline is also strong because the one-head matrices are small enough to fit comfortably in cache. A more favorable FPGA comparison would require either more parallel heads, larger batches, longer sequences, or a more integrated FPGA-native attention kernel that keeps the full computation on card.

Another bottleneck is that the current design does not use

all available HBM parallelism. It maps logical buffers across several banks, which prevents all traffic from contending for one memory channel, but it does not fully exploit the 32 HBM pseudo-channels available on the U55C. The buffer sizes in the verified one-head tests are also small compared with total HBM capacity. Thus, the design is not capacity-limited; it is limited by orchestration, launch granularity, and the amount of data staged between kernels.

The current design also accelerates only one selected head. TinyLlama has 32 query heads and 4 key/value heads. In the real model, multiple heads are computed as part of the same layer. An FPGA design that processes several heads or several Q chunks concurrently could use more DSPs and HBM bandwidth and would provide a more realistic throughput comparison. This project stops before that scale-out step so the one-head path can be verified carefully first.

E. Optimization Direction

The optimization checklist identifies three architectural improvements before minor clock or report-guidance tuning. First, full Q/K/V and intermediate buffers should remain resident on device so the host reads back only final attention output during normal operation. Second, launch count should be reduced by using larger-grain kernels that process all K chunks for one Q chunk. Third, online softmax should be fused with value accumulation so the design no longer materializes the full $S \times S$ probability matrix. In online softmax, each query row maintains a running maximum, exponential sum, and 64-element value accumulator while streaming through K/V chunks. This would replace the current logit-materialization, full-row softmax, probability staging, and V weighted-sum sequence with a single attention-output-producing kernel per Q chunk.

The expected optimization path is incremental. A device-resident flow should first copy Q, K, and V to device memory once and read back only final attention output in the normal path. A launch-count reduction should then move from per-tile launches to one pre-softmax and one value-accumulation launch per Q chunk. Finally, online softmax should remove the need to store a full $S \times S$ probability matrix. Each step should preserve the existing reference checks so that performance changes do not obscure correctness.

F. Limitations

Several limitations should be stated explicitly. The design is not full TinyLlama inference and does not report model-level tokens per second. The real TinyLlama saved-vector coverage is limited to $S = 16$ and $S = 64$ in the verified master result, while larger $S = 128, 256, 512$ tests are synthetic. The GPU baseline was measured on a separate laptop rather than the U55C host machine. Finally, the FPGA implementation uses FP32 for softmax and V accumulation, which simplifies verification but is not necessarily the final numeric format for a production accelerator.

Another limitation is that the design does not yet use a KV-cache-aware decode path. This matters because LLM

inference has two phases. During prompt prefill, many query rows are processed together. During decode, each generated token usually introduces one new query row that attends over cached keys and values. The current full-sequence saved-vector tests are closer to prefill validation. A deployed edge LLM accelerator would need a cache layout, update policy, and decode-time host or controller interface. Those pieces are outside the verified scope but are natural extensions once the attention kernel itself is more integrated.

VI. CONCLUSION

This project implemented and verified an isolated one-head transformer attention block on a Xilinx Alveo U55C FPGA. The final fused design computes INT8 query-key scores, applies causal masking and dequantization scaling, performs full-row softmax, and computes $\text{softmax}(QK^T)V$ to produce FP32 one-head attention output. The design is verified through local reference checks, HLS simulation and synthesis, hardware emulation, and real U55C execution. Synthetic sequences up to $S = 512$ and TinyLlama-derived saved vectors at $S = 16$ and $S = 64$ pass final output verification.

The performance results are honest: the FPGA implementation is correct and timing-clean, but it is not faster than the CPU or GPU baselines for the current isolated one-head workload. The fused pre-softmax kernel improves the staged FPGA baseline, yet the design remains dominated by many small kernel launches and HBM-visible intermediate transfers. Therefore, the project should be viewed as a validated hardware baseline and design study rather than a final high-performance LLM accelerator. The next step is to make the computation more FPGA-native by keeping buffers resident, reducing launch count, fusing online softmax with V accumulation, and eventually scaling across multiple Q blocks or attention heads.

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VII. STUDENT CONTRIBUTIONS

Om designed and implemented Track A Steps 1 and 2. This included the 16-way unrolled INT8 score GEMM work, 64-bit

packed memory I/O for the score path, and the fused causal-mask and score-scale kernel that removed an intermediate HBM round trip. Om also ran HLS synthesis and helped verify the early optimized score path on U55C hardware.

Samarth designed and implemented Track A Steps 3 through 5. This included the Python full-sequence tiling reference, full-row softmax support for $S \leq 512$, the XRT host tiling loop, Pass 1 host double buffering, and the fused score-mask-scale kernel used in the final frozen design. Samarth also verified tiled synthetic runs on the real U55C for $S = 8, 64, 128, 256, 512$.

Rijul Wankhade designed and implemented Track B. Rijul added the `v_weighted_sum` HLS kernel for the `softmax@V` stage, integrated it into the host flow as the value-accumulation pass, and changed verification so the FPGA checks final one-head `attn_out` rather than only attention weights. This completed the one-head attention output path.

Satyarth implemented Track C and Track D. This included TinyLlama Q/K/V extraction through a PyTorch forward hook, INT8 symmetric quantization for Q and K, full-sequence real-vector export at $S = 16$ and $S = 64$, CPU and GPU baseline benchmarks, and the performance comparison tables and analysis used in this report. Satyarth also verified the complete path with real TinyLlama-derived vectors on the U55C.